

Unit 5: Jaccard distance/dissimilarity calculations

The purpose of this exercise was to calculate the Jaccard distance between individuals (Jack, Mary and Jim) using binary pathological test results, provided by University of Essex (2026). Jaccard distance measures how dissimilar two sets are by comparing only the presence of characteristics whilst ignoring shared absences (Noble, n.d.).

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	A
Mary	F	Y	N	P	A	P	N
Jim	M	Y	P	N	N	N	A

Calculations were carried out where M,Y,P and A = 1 (present) and where F and N = 0 (absent). The Jaccard distance was calculated as Distance = 1 – J.

$$Jaccard = \frac{f_{01} + f_{10}}{f_{01} + f_{10} + f_{11}}$$

The results of this calculation showed that Jack and Jim are the most similar, with the smallest Jaccard distance (0.400). Jim and Mary are the most dissimilar, with the largest Jaccard distance (0.857).

Pair	Jaccard Similarity	Jaccard Distance
Jack – Mary	0.333	0.667
Jack – Jim	0.600	0.400
Jim – Mary	0.143	0.857

This demonstrated how Jaccard distance can identify similarities between patients based on shared clinical characteristics while ignoring characteristics that are absent in both patients.

The exercise helped me to understand similarity and distance measures used in machine learning. I learned that Jaccard distance is particularly suitable for binary datasets as it focuses on the presence of important features rather than shared absences. For healthcare datasets, this approach can help to compare patients based on their symptoms or test results, supporting clustering and classification tasks. It also highlighted the importance of selecting an appropriate similarity measure depending on the type of data being analysed.

References

Noble, J. (n.d.) *What is Jaccard similarity?* IBM Think. Available at: <https://www.ibm.com/think/topics/jaccard-similarity> (Accessed: 20 July 2026).

University of Essex, (2026) *e-Portfolio Activity: Jaccard Distance/Dissimilarity Calculations*. Machine Learning. Available at: <https://www.my-course.co.uk/mod/page/view.php?id=1344683> (Accessed: 20 July 2026).